

Project Report: Information Retrieval Engine

Introduction:

- 1) As part of the project, an Information Retrieval (IR) engine is to be built that makes forward and inverted
- 2) The IR engine will ingest documents and provide a means to index and retrieve those documents based
- 3) The primary components are ForwardIndex, InvertedIndex, TextParser, Retriever, and Main.

Design and Implementation:

Forward Index:

- 1) The ForwardIndex class keeps a mapping of document IDs to word frequencies.
- 2) Each document is associated with a unique identification number and stores the words in that document
- 3) Method addDocument adds a document to the forward index.
- 4) The method saveToFile saves the index to a file.

Inverted Index:

- 1)The InvertedIndex class keeps a mapping of words to document ids and frequencies of that word in the c
- 2) This index allows efficient retrieval of documents that include specific terms.
- 3)The method addDocument adds a document to the inverted index.
- 4)The method saveToFile saves the index to a file.
- 5)The method search retrieves documents and frequencies for a given term.

Retrieval:

- 1)The Retriever class uses the inverted index to retrieve documents based on search terms.
- 2)The method retrieve returns a map of the document ID and frequency of the document for that specific search term.

Main Class:

- 1)The Main class orchestrates the entire process, from loading stop words, parsing documents, and building the indices.
- 2)The program reads the documents from the specified folder, processes them, and builds the forward and inverted indices.
- 3)The program prompts the user for a search term and displays the documents and their frequencies for that term.

Steps of Execution

Loading Stop Words:

- 1) Stop words are loaded from the file (stopwordlist.txt) into a set.
- 2) Initializing Components: The Porter stemmer and the text parser are initialized.
- 3) The forward and inverted indices are initialized.
- 4) Processing Documents: The documents are read from the directory './documents'.
- 5) Each document is parsed to extract the words that are added to the forward and inverted indices.
- 6) Saving Indices: The indices are saved to files - forward_index.txt for forward index and inverted_index.txt for inverted index.
- 7) Search: The program prompts the user for a search term.
- 8) The retriever will search the inverted index ((which is the index for retrieval) for the search term and display the results.